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REVIEW ARTICLE

Visual servoing in medical robotics: a survey. Part I: endoscopic and direct vision imaging – techniques and applications

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Abstract

Background Intra-operative imaging is widely used to provide visual feedback to a clinician when he/she performs a procedure. In visual servoing, surgical instruments and parts of tissue/body are tracked by processing the acquired images. This information is then used within a control loop to manoeuvre a robotic manipulator during a procedure.

Methods A comprehensive search of electronic databases was completed for the period 2000–2013 to provide a survey of the visual servoing applications in medical robotics. The focus is on medical applications where image-based tracking is used for closed-loop control of a robotic system.

Results Detailed classification and comparative study of various contributions in visual servoing using endoscopic or direct visual images are presented and summarized in tables and diagrams.

Conclusion The main challenges in using visual servoing for medical robotic applications are identified and potential future directions are suggested. 'Supervised automation of medical robotics' is found to be a major trend in this field. Copyright © 2013 John Wiley & Sons, Ltd.

Keywords visual servoing; medical robotics; medical imaging; minimally invasive therapy; image-based tracking

Introduction

New developments in robotic technology have been applied to diagnosis, therapy, and surgery. Minimally invasive therapy is one of the emerging applications of medical robotics, which has been the focus of research for more than two decades. A major challenge in closed-loop control of robot manipulators is providing accurate position feedback of the robotic tool. This becomes more critical when the robotic tool is inserted into the human body, as well as when flexible tools, such as catheters, needles and endoscopes, are used. There are several techniques for fast and accurate measurement and control of the position/orientation of a robotic tool, including: (a) feed-forward, model-based control using robot kinematics; (b) electromagnetic (EM) tracking; (c) image-based tracking, using medical imaging modalities, and other techniques which are less commonly used. Model-based tracking cannot be used for flexible robots and is prone to errors, due to modelling mismatches and noise or error in encoder/resolver readings. Therefore, it is not appropriate as a versatile solution on its own. EM trackers are very sensitive to electromagnetic interference and are

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affected by metallic tools, which limits their usefulness in clinical applications. In addition, mounting EM trackers on a distal part of a tool that is inserted into the human body is a challenging task. Medical imaging modalities are commonly used by clinicians to obtain feedback, both pre-operatively and intra-operatively. Applying image-processing algorithms to existing imaging modalities is an efficient technique for providing fast and accurate position feedback that can be used to control robot manipulators. 'Visual servoing' is the common term used for the application of visual feedback in robot control (1).

Troccaz *et al.* (2) provide an in-depth review of medical robotics and the underlying technology. Detailed reviews of visual servoing for industrial and service robotic manipulation have been carried out by Hutchinson *et al.* (1), Corke (3) and Kragic and Christensen (4). In the medical context, Groeger *et al.* (5) reviewed motion-tracking methods using endoscopic images that can be employed for visual servoing in minimally invasive surgery. In this paper, the definition of visual servoing in medical robotics is generalized to include various types of imaging modalities to provide visual feedback. However, only some of the image-guided medical robotic systems use visual servoing. In fact, a majority of such systems close the control loop through a human operator. A detailed review of such techniques has been given by Peters (6). On the other hand, not all automatic/semi-automatic medical robotic systems use visual servoing, e.g. other types of position sensors, such as EM and optical (such as laser and infrared) trackers are also used for closed-loop control of medical robots.

Visual servoing has been widely used for different robotic systems and has received more attention in medical robotic applications in the last decade. Uecker *et al.* (7) and Wei *et al.* (8) were among the first to report applications of visual servoing in medical robotics using the AESOP[1] robot and endoscope images for laparoscopic surgery. Following this pioneering work, there have been several research contributions on visually servoed medical robots. This paper provides a review of the developments in this field by covering most of the relevant research activities in medical robotics which involve visual servoing using optical vision.

Focus and outline

To clarify the extent of the work covered in this paper, we consider a system to be 'visually servoed' only if the imaging information is used for closed-loop control/guidance of a robotic system in real time. It should be noted that image-guided medical applications without direct use of imaging in the robot control loop are not the focus of this paper. This also includes applications where medical imaging is only used for pre-operative and/or intra-operative registration, calibration and/or motion planning of robotic systems.

The rest of the paper is organized as follows. The subjects are categorized based on the imaging modality and their medical applications. Tomographic imaging modalities are covered in Part II of the paper. For each case, image-based tracking and visual servoing techniques are briefly described, followed by a short review of the results. This paper also presents a brief overview of existing visual servoing configurations and controllers as well as a short review of conventional software packages and processing units used in visual servoing systems. The paper concludes with an overall view of the current situation and possible future directions in this field of research. Wherever possible, tables and diagrams are included to help in classifying and comparing different contributions in the area.

Materials and methods

A common and important feature in visual servoing techniques for medical robotics is that real-time imaging should be fast enough to avoid any aliasing in image-based tracking and also to adapt to fast transients in visual servoing control. This limits the application of certain types of imaging modalities which cannot provide images in real time. The imaging modalities used for visual servoing and reported in the literature include: ultrasound (US) imaging; X-ray fluoroscopy; magnetic resonance imaging (MRI); computed tomography (CT) fluoroscopy; endoscopic imaging; and direct vision. This section presents the state of the art in visual servoing techniques that use either endoscopes, external cameras or microscopes to provide image feedback. For a review of current techniques for other imaging modalities, refer to Part II.

Endoscopic imaging

Endoscopic imaging is widely used to provide visual feedback to the clinician while surgical instruments are inserted through small incisions into the patient's body. Endoscopes are also widely used for diagnostic and treatment purposes while being inserted through natural openings of the body. Groeger *et al.* (5) have studied different motion-tracking techniques using endoscopic images in minimally invasive surgical applications. Several different visual servoing applications using endoscopic images are discussed next.

Laparoscopy

Endoscopic imaging can be used in visually servoed laparoscopy in order to control the position of the camera (e.g. to always keep the tools or a section of an organ in the field of view) or to control the movement of laparoscopic tools.

[1]Automated Endoscopic System for Optimal Positioning (AESOP), manufactured by Computer Motion Inc., was the first surgical robot approved by the US Food and Drug Administration (FDA). Computer Motion Inc. was acquired by Intuitive Surgical in 2003.

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In one of the first studies in this field, Uecker *et al*. (7) developed a system for automatic positioning of a laparoscopic camera using an AESOP robot, by tracking the laparoscopic tools and using a visual servoing controller. A four-step feature detection and tracking algorithm was developed to track a laparoscopic tool at a rate of 15 fps, and a Jacobian inverse controller was developed to control the AESOP robot in order to adjust the laparoscopic camera to keep the laparoscopic tool in the field of view.

Voros *et al*. (9) reported a lightweight sterilizable body-mounted robot with three degrees of freedom (DOFs), called the ViKY Robotic Scope Holder, which can act as a third hand for the surgeon by holding the endoscope. The system provides near real-time detection and tracking of the instruments. In this work, first the surgeon indicates the point of insertion of the tool on the image. Next, by knowing the insertion point, the edges of the tool are detected by image processing. Finally, the instrument's position is inputted into a visual servoing loop in order to automatically move the endoscope toward the tip of the instrument. The system was evaluated on cadavers through several procedures. The operation time was comparable to cases in which the endoscope was controlled manually. ViKY obtained CE mark in 2007 and FDA clearance in 2008 (10) and is currently commercialized by EndoControl (Grenoble, France).[2]

Bourger *et al*. (11) used a combination of efficient second-order minimization (ESM) tracking and histogram matching in order to track a region of interest even when a major part of the image was occluded. The method used colour images and the results have been applied to visual servoing of an AESOP robot on a maquette of internal organs. The method can be applied for laparoscopic abdominal surgery where different organs and surgical tools can occlude target areas, and was used to track a user specified area.

Krupa *et al*. (12) used a laser pattern projected on the organ to track the organ. They also attached active markers on the surgical tool. By tracking the organ and tool in the endoscopic images, they developed a visual servoing controller for 3-DOF positioning of a surgical tool with respect to the tissue. Krupa *et al*. (13) used the same technique for automating subtasks in laparoscopic surgery of the liver. They combined visual servoing with force control to maintain a remote centre of motion at the point of insertion (i.e. through a trocar) while performing repetitive tasks autonomously, such as cleaning and suction.

Kim *et al*. (14) developed a real-time visual tracking algorithm for visual servoing in laparoscopic surgery. A two-stage adaptive conditional density propagation algorithm was proposed in order to detect accurate position of the laparoscopic tools from endoscopic images. This method is a sampling-based algorithm. The sampled positions of the laparoscopic tools and their colour features at each position are used for this purpose. The main advantage of this method is its adaptability to gradual or rapid illumination changes, as well as varied surgical situations (e.g. overlapping tools).

Osa *et al*. (15) used visual servoing in order to develop a semi-autonomous robotic laparoscopic system. This system consists of four ceiling-mounted 6-DOF Mitsubishi MELFA arms holding the laparoscopic tools and an endoscopic camera. Colour markers are attached to the tool shafts in order to obtain the two-dimensional (2D) positions of the tools in the endoscopic view. Image-based visual servoing (IBVS) is applied to control the linear velocity of the robot arms, using endoscopic images. In addition, a geometric constraint is employed in order to keep the tools stationary at the point of insertion, which determines the angular velocity of the tools. The positioning error was reported to be less than the measurement resolution (i.e. < 1 mm).

Automatic suturing

Endoscope-based visual servoing can also be used in suturing, either by assisting the alignment of the needle with the point of insertion, or by controlling the motion of laparoscopic tools, e.g. in tying a knot. Staub *et al*. (16) developed a system for image-guided tissue stitching. They used a laser pointer to specify the point of stitching. This point is detected by the image-processing algorithms through a stereo-endoscopic system. The stereo-endoscope is then used for automatic alignment of curved needles with the specified points through a visual servoing controller to perform automatic suturing.

Hynes *et al*. (17) developed an uncalibrated visual-servo system for positioning pairs of minimally invasive surgical tools during suturing, using a dual-arm robot. In this study, the effects of uncertainty of camera positioning and mounting of the surgical instrumentation were removed through uncalibrated servoing and a kinematic model of the robot arms. A combination of template matching (i.e. normalized cross-correlation) and colour patches on the tools allowed good tracking of the tool tips at a rate of 25 fps. An inverse Jacobian controller was used to control the motion of the tools.

Padoy and Hager (18) formulated the tracking problem of thin deformable objects as a three-dimensional (3D) non-uniform rational B-splines (NURBS) estimation problem and developed a technique for real-time tracking of threads from stereo-endoscopic images. Using this formulation, they developed a visual servoing technique, using the daVinci® surgical robot, to perform automatic cutting of sutures using the robot's third arm while the surgeon held the thread via the other robotic arms (19). Rates of up to 5 fps were reported. They also developed a human–machine collaborative control approach, in which a surgical task (suturing in this case) is divided into manual and automatic subtasks. Upon completion of each manual subtask, the system performs the learned subtasks autonomously, but under the surgeon's supervision (20).

Iyer *et al*. (21) proposed a single-arm robotic system with visual feedback to perform suturing tasks semi-

[2]http://www.endocontrol-medical.com

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autonomously. A 6-DOF robotic manipulator steers a curved suturing needle from the entry point to the exit point (defined by the surgeon), using the endoscope. The system was tested on a fixed tissue pad phantom. This system does not implement a real-time visual servoing controller. The robot is stopped; the needle is detected and the robot is moved to pick the needle, based on pre-calibration information.

Diagnostics

Reilink *et al*. (22) developed image-processing algorithms for guiding a 2-DOF steerable endoscope in order to control its tip orientation towards the direction of the lumen. Two different image-processing techniques were used: (a) an optical flow-based technique similar to what is used for corridor following for mobile robots; and (b) an intensity-based image processing technique to find the centre of the dark region representing the gastrointestinal tract. The results were validated through simulation and also using a gastroscope on an anatomical model. An overall improvement of 68% was reported compared to manual steering. The proposed technique is an image-based visual servoing (IBVS) approach.

Natural orifice transluminal endoscopic surgery (NOTES)

Endoscope-based visual servoing is also beneficial in robot-assisted interventions through natural openings of the body. Ott *et al*. (23) motorized an endoscope in order to compensate for respiratory motion in NOTES applications. They modelled the non-linearities of the endoscope actuation mechanism (mainly backlash and hysteresis). They deployed a prototype repetitive controller and a repetitive generalized predictive controller (R-GPC) scheme to reject the periodic motion through visual servoing. Displacements in the image were estimated through ESM tracking. Although this method was able to reject periodic disturbances caused by respiration, it had poor response to non-periodic motion, which is caused by insertion/retraction of the endoscope by a human operator. Ott *et al*. (24) approached this problem by using a switching controller for model-based decoupling of reference tracking and periodic disturbance rejection in a prototype repetitive controller (PRC) control scheme.

Dumpert *et al*. (25) developed a miniature visually-servoed robotic platform which emulated laparoscopic surgery for NOTES procedures. The body of the robot contains a stereo camera pair, which acts as an overhead camera viewing its two arms. Colour markers on the robot arms are used for position tracking. A simple proportional–integral (PI) controller is employed to control the robot arms. The maximum measurement error during tracking was 0.7 mm, with a mean error of 0.3 mm. However, the experimental results demonstrated that the system is prone to occlusion problems in *in vivo* situations.

Direct vision

Direct camera vision can also be used for visual servoing in medical robotic applications wherever a direct line of sight and a clear workspace exist between the camera and the object to be tracked during open surgery. Direct vision is also used to simulate other imaging modalities. Some visual servoing applications using direct vision are discussed in this section.

Kallem and Cowan (26) studied the visual servoing of a steerable flexible needle using stereo camera vision. The stereo vision and a transparent tissue phantom provided an experimental set-up which represented bi-plane fluoroscopy (or 3D ultrasound) with real human tissue. A non-holonomic steering model of a bevel-tip needle was used to design a model-based observer and a feedback controller to guide the needle to a desired planar slice of tissue, while the operator controlled the insertion speed. Webster *et al*. (27) used a similar method for visual servoing of concentric-tube continuum robots (active cannula).

Nishikawa *et al*. (28) used the Polaris[3] optical tracking system for servoing laparoscopic tools. Busack *et al*. (29) and Stein *et al*. (30) studied the application of optical tracking using the ART[4] optical tracking system, for laser osteotomy with respiratory motion compensation.

C-arm positioning

Navab *et al*. (31) used a camera-augmented mobile C-arm (CAMC) for repositioning and reorientating a C-arm without the need for additional exposure. Passive markers were installed on the patient's body and tracked using vision. Two applications were discussed in the paper. (a) One application was the so-called 'down-the-beam' positioning of the C-arm when it is required to insert a linear surgical tool, such as a needle, to a deep anatomical target, or to position a screw (e.g. a transpedicular screw) in an interlocking hole (e.g. in spinal surgery). In this application, pre-operative CT was used to define the insertion axis. This was then transformed to the C-arm coordinates and position-based visual servoing was used to bring the C-arm towards the desired position. Image-based visual servoing was then used for optimal positioning of the C-arm close to the target position. The target image was generated from the CT images. (b) The second application was intraoperative repositioning of a C-arm, where pure image-based visual servoing was used to reposition the C-arm at its previously known position. The actuation commands generated by the visual servoing algorithm were followed by a human operator, although it could also be done automatically using a motorized C-arm.

[3]http://www.ndigital.com/medical/polarisfamily.php
[4]http://www.ar-tracking.de

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Orthopaedic surgery

In orthopaedic surgery, skill is needed to manipulate a surgical tool accurately and to reach an expected position/orientation on a target bone. A medical robot is considered to be helpful in meeting such demands for improving accuracy, as well as reducing surgeon fatigue. A drawback of conventional robot-assisted orthopaedic surgery is the necessary invasive clamping of anatomical objects. This is done to maintain the spatial alignment to the robot. This fixation can lead to lesions in soft and hard tissue. Furthermore, if there is any unexpected movement in the robot workspace, this spatial alignment has to be renewed.

To avoid such invasive clamping, Ranftl *et al*. (32) developed a hybrid force/velocity control approach for implementation in a semi-active strategy for total knee arthroplasty. In this method, an industrial robot equipped with a vision system performed the machining tasks. With the help of dynamic registration, the vision system fed back any unexpected motions in the workspace to the robot controller, allowing the device to compensate autonomously for such changes. Similarly, by taking advantage of a stereo vision system, Chen *et al*. (33) developed an unconstrained virtual bone clamper for total knee replacement surgery. Several markers were placed on the robot and the bone. The stereo camera tracked these markers. Then an end-point, closed-loop control strategy was used in order to modify the robot trajectory, based on the change in the location of the bone. It was reported that the proposed modified endpoint closed-loop visual servo technique can achieve positioning accuracy similar to that of the traditional fixed bone method.

Open heart surgery

Although it is not currently possible to use vision in interventional cardiology, due to the difficulties of vision through blood and lack of direct line of sight, visually guided procedures can be used in cardiac surgery using endoscopes in laparoscopic cardiac surgery and normal vision in open heart surgery.

Bachta *et al*. (34) explored visual servoing techniques for beating-heart coronary artery bypass grafting (CABG). Salcudean *et al*. (35) developed a system that moved a platform supporting a surgeon's hand, based on visual feedback from the beating heart to compensate for heart beat motion and reduce tremor. Nakamura *et al*. (36) used a high-speed camera (955 fps) to track a marker on the heart surface. They used the tracking data to compensate for the heart-beat motion using a 4-DOF slave robot. They also used tracking to stabilize the visual feedback to the user. The user controlled the slave robot through a master–slave control scheme in such a way that he/she was operating on a fixed heart. Image-based visual servoing (IBVS) was used to stabilize visual feedback and a position-based visual servoing (PBVS) technique was used to control the surgical tool.

Bachta *et al*. (37) developed a system called Cardiolock, which used visual servoing to compensate for the residual heart-beat motion after a cardiac stabilizer had been used. A high-speed camera was used to track the bending of a compliant tool at a rate of 333 Hz and piezo-actuators were used to compensate for this. An H_∞ controller was used to reject the heart-beat perturbations. Using this controller, a targeted accuracy of 0.1 mm was obtained in *in vivo* porcine models (38).

Ginhoux *et al*. (40,41) developed a tracking mechanism for cardiac surgery on a beating heart. They used multiple active markers [light-emitting diodes (LEDs)] mounted on the surface of the heart and a high-speed camera to capture images and track the markers at a rate of 500 Hz. They used a generalized predictive controller (GPC) with an internal model of the heart (39), and also a repetitive GPC (40), for prediction. The motion was tracked using 2D images and visual servoing was used to compensate for the beating heart motion.

Blankenstein (41) studied the application of high-speed visual servoing for physiological motion compensation by taking into account the dynamics of imaging due to visual temporal sampling. Active LED markers were used to track the tool at rates of up to 500 Hz. Visual servoing of simple 1-DOF rotational and linear motors was tested at rates of 100 and 500 Hz.

Microsurgery

New developments in microrobotics has enabled surgeons to perform delicate procedures with higher accuracy. To this end, visual servoing (based on stereo microscopes) can play an important role in microsurgery, through improving accuracy of procedures and reducing hand tremor. Becker *et al*. (42) developed a tremor-reduction system for microsurgery. A stereo microscope ($\times$25 magnification) was used to track the tip of a hand-held microsurgery tool and visual servoing was used to assist the operator to reach a target with an accuracy of ca. 17.5 μm. This system was used for laser retinal surgery (43) and retinal vessel cannulation (44). The concept of virtual fixtures was used to assist in following a certain trajectory in microscope images, while tremor cancellation and motion scaling were performed on the hand-held device (45).

Kahrs *et al*. (46) used visual servoing for laser-based bone removal in the inner ear (cochleostomy), using a $\times$7 magnification objective lens. Yesin *et al*. (47) developed a system for intraocular microsurgery using magnetically driven microrobots. They developed an active defocused tracking technique to estimate the 3D position of the microrobots in the vitreous humor (the gel-like substance between retina and the lens in an eye-ball), using 2D microscope images. A visual servoing technique was used to move the microscope along the view axis in order to keep a microrobot in a slightly defocused position in the microscope images. Simple thresholding and centroid detection provided tracking of the microrobots in the 2D microscope images and visual servoing of the

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microrobots through magnetic fields. The visual servoing controllers were implemented at a rate of 60 Hz.

Zong *et al*. (48) developed a visual servoing method for suturing in microsurgical keratoplasty (i.e. corneal grafting). To this end, they took advantage of a purpose-made microrobotic suturing end-effector, which was attached to a 6-DOF modular robot arm. Two CCD cameras were mounted on a surgical microscope. Visual servo control was applied for guiding a needle to reach the penetrating target point. In this system, the control values were computed directly on the basis of image features, subject to accuracy, stability and robustness.

Applications

To have a classification of the literature in the field of visual servoing, we have listed various contributions based on their applications in separate tables. Each table includes information in order to compare the results reported by the papers. Some of the categories that have been the subject of research on visual servoing with a significant number of contributions are listed as follows: (a) physiological motion compensation (Table 1); (b) visual servoing of the imaging device (Table 2); (c) visual servoing of the surgical tool (49) vs servoing of the patients (50); and (d) visual servoing for tremor compensation (42). Table 3 provides a list of different research contributions based on their clinical applications.

An analysis of the medical applications of the reported contributions shows that only a small minority have focused on applications dealing with rigid bodies (e.g. in orthopaedics). The reason is that a conventional image-guided approach with accurate registration, such as StealthStation® Treatment Guidance System (Medtronic), would suffice for most of such applications. A majority of the reported applications of visual servoing focus on cases in which a dynamic element is involved. This includes soft tissue deformations, physiological motion of organs due to heart-beat and respiration and the operator's unintentional motions, such as tremor.

From Table 3, one can see that most of the visual servoing applications use optical vision in terms of normal or endoscopic cameras or microscopes. The second most-used modality is ultrasound imaging, which is discussed in Part II of this paper along with other tomographic imaging modalities.

Results

Beside the problem of having direct access to image data in real time [see Part II of the paper], other main challenges in using visual servoing for medical robotic applications are as follows:

- *Robustness and reliability of image-based detection and tracking algorithms*. Yilmaz *et al*. (51) provide a comprehensive survey of image-based object-tracking techniques and, although they do not cover medical imaging, they do cover similar techniques and problems. Several robust tracking techniques have been reported but all have assumptions that make the tracking problem tractable. Smoothness of motion, no major occlusions, illumination constancy and high contrast of object with respect to its background are assumed in most cases. Such conditions are not generally valid in medical applications when visual images are used to track an object. It can be argued that more reliable techniques are more computationally intensive and therefore availability of high-end computation platforms could potentially solve part of the problem. This challenge is also discussed in Future directions of visual servoing techniques.
- *Image resolution and tracking accuracy*. Higher image resolution can result in higher tracking accuracy. With advances in medical imaging technology, high-quality images are becoming available. As an example, high-definition digital endoscopic images have replaced standard analogue video qualities over the last decade.
- *Image-capture rate*. The image-capture rate is limited by the imaging modality. It is commonly understood that, because a human being (in this case a surgeon) does most of his/her physical tasks via visual servoing, it can be argued that the capture rate of a human eye should be sufficient to perform a surgical task. On the other hand, reliability and accuracy are considered

Table 1. Visual servoing techniques used for physiological motion compensation in medical robotic applications

Contribution	Imaging modality	Markers	Rate (Hz)	DOFs	Motion	Tracking	Controller	Application
Ginhoux *et al*. (39)	HS-vision	Active	500	2	H	–	GPC	General cardiac
Ginhoux *et al*. (40)	HS-vision Endoscope	Active	500 25	2	H-R	–	R-GPC, adaptive GPC	General cardiac
Nakamura *et al*. (36)	HS-vision	Passive	955	4	H	–	See the section on Open heart surgery	CABG
Bachta *et al*. (37)	HS-vision	None	333	1	H	–	H_∞	CABG
Ott *et al*. (23)	Endoscope	None	25	2	R	MST, ESM	R-GPC, PRC	NOTES
Ott *et al*. (24)	Endoscope	None	25	2	R	ESM	Customized PRC	NOTES
Ranftl *et al*. (32)	Vision	None	10	6	Various	Template matching	Hybrid force/velocity	Orthopaedic surgery
Chen *et al*. (33)	Vision	Passive	10	7	Various	Template matching	Hybrid force/velocity	Orthopaedic surgery

Column 5 shows the number of DOFs used for actuation; column 6 lists the source of physiological motion to be compensated for, heart beat (H) and respiration (R). The tracking method is not reported for the techniques that use markers and/or HS-vision; it is generally a simple and fast segmentation technique (such as thresholding).

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Table 2. Visual servoing of an imaging device

Contribution	Imaging modality	Markers	Rate (Hz)	DOFs	Tracking/control	Application
Navab *et al*. (31)	Vision/XRF	Passive	Varying	5	–	C-arm (re)positioning
Zhang and Payandeh (69)	Endoscope	Passive	≤ 30	3	–	Laparoscopic surgery
Ott *et al*. (23,24)	Endoscope	No	25	2	MST, ESM/PRC, R-GPC	Respiratory motion compensation
Reilink *et al*. (22)	Endoscope	No	≤ 30	2	See the section on Diagnostics with Endoscopic imaging	Gastroscopy
Voros *et al*. (9)	Endoscope	No	–	3	Edge detection, landmarks	Laparoscopic surgery
Yesin *et al*. (47)	Vision	No	60	3	Thresholding, centroid detection	Intraocular microsurgery

Table 3. Classification of visual servoing techniques in medical robotics based on their applications

Application	Contributions	Modality	Tool	DOFs	Rate (Hz)	Configuration	Trials
General cardiac surgery	Salcudean *et al*. (35)	Vision	Hand support	–	NA	B	–
CABG	Kennedy and Desai (55)	Vision	Robot end-effector	3(7)	30	C	Phantom
	Bachta *et al*. (38)	Vision	Hand support	–	333	B	
Cochleostomy	Kahrs *et al*. (46)	Vision	Laser ablation electrode	–	–	D	Phantom
NOTES, gastroscopy	Reilink *et al*. (22)	EN	Endoscope	2	≤ 30	E	Phantom
	Ott *et al*. (23)	EN	Endoscope	2	25	B	*In vivo*
	Ott *et al*. (24)	EN	Endoscope	2	25	B	Phantom
	Dumpert *et al*. (25)	Vision	Miniature robot	4	5	D	Benchtop
Laparoscopic surgery	Padoy and Hager (19)	EN	Suture thread	7	5	E	Phantom
	Iyer *et al*. (21)	EN	Suture needle	6	–	E	Phantom
	Nageotte *et al*. (70)	EN	Suture needle	4	20	D	Phantom
	Zhang and Payandeh (69)	EN	Endoscope	3	≤ 30	D	Benchtop
	Uecker *et al*. (7)	EN	Laparoscopic instrument	3	15	D	Simulation
	Krupa *et al*. (12)	EN	Laparoscopic instrument	3	50	D	*In vivo*
	Kim *et al*. (14)	EN	Laparoscopic instrument	3	–	D	Benchtop
	Hynes *et al*. (17,71)	EN	Laparoscopic instrument	3	25	D	Phantom
Proton therapy	Belaroussi and Morel (50)	Vision	Table	6 (6)	200 ms	F	Phantom
Microsurgery	Zong *et al*. (48)	Microscope	Miniature robot	7	–	D	Phantom
	Becker *et al*. (42)	Microscope	Hand-held robot	3	12	B	Benchtop
	Yesin *et al*. (47)	Microscope	Micro-robot	3	60	B	Benchtop
Orthopaedics	Ranftl *et al*. (32)	Vision	Industrial robot	6	10	D	Phantom
	Chen *et al*. (33)	Vision	Robot and drilling tools	7	10	D	Phantom
Teleoperation	Kennedy and Desai (55)	Vision	Rod	1 (6)	30	D	Phantom

The active DOF used for visual servoing are reported in column 5; total number of active DOFs are listed in round brackets. Column 4 specifies the tool that is visually servoed; the term 'rod' is used to denote any long cylindrical rigid tube used, such as a grasper, scissors, cutter, needle driver, clip deployment device, etc. The servo rate is reported in the last column in terms of either frequency (Hz) or time period (ms). Column 7 (Configuration) refers to the visual servoing configurations introduced in the section on visual servoing configuration.

significantly more important than speed in performing a medical intervention. Therefore, we can conclude that existing image-capture rates are not the major limiting factor in implementing a visually servoed system, although a control system can always benefit from higher sensor rates when the actuation rates are higher.

- *Effect of latency caused by the time required for image capture and transfer, and processing in the control loop*. The overall latency in a control loop is an important design criterion and high latency can cause instability. The latency in image capture is dependent on the imaging modality. Technological improvements are expected to reduce this effect. In certain modalities, such as US imaging, there is a physics-based limitation (e.g. limited sound speed). Image transfer rates are also limited due to frame rates, image quality and the available bandwidth. There can be a trade-off based on the requirements of a specific application. Latency due to image processing depends directly on the computational efficiency of the applied processing algorithms, which is discussed next.
- *Computational efficiency of the image-processing algorithms*. A more accurate and robust algorithm usually comes at the price of higher computational cost. We have briefly discussed the effect of the increasing processing power of CPUs and GPUs on this problem in Future directions.
- *Offline calibration and registration*. There are visual servoing techniques which can be applied to uncalibrated systems (17), but some initial calibration or registration can potentially increase the efficiency and improve the convergence of a visually servoed system. Therefore, a majority of the reported cases have used some type of calibration and/or registration. Adding a calibration/registration step into the clinical workflow might become challenging in some cases.

A survey of the literature shows that endoscopic and ultrasound imaging [discussed in Part II of the paper] are the most popular imaging modalities for visual servoing. The main reason behind this is that visual servoing requires continuous imaging. This can significantly increase the exposure of a patient to harmful radiations when using X-ray or CT fluoroscopy, although they provide a better contrast compared to US imaging

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and can see through organs, as compared with endoscopic imaging. The clinical availability and relatively lower costs of these modalities also makes them better candidates for visual servoing applications. This comes at the price of more complicated processing, due to the poor quality of US images (see part II of the paper) and the limitation of endoscopic images, such as the requirement for line of sight, and sensitivity to illumination changes.

Discussion

Different visual servoing techniques have been used in medical robotic applications. Instead of using the classic categories of visual servoing, such as IBVS, PBVS and 2.5D visual servoing (4), or the taxonomy used by Sanderson and Weiss (52), the techniques in the cited contributions are listed based on the visual servo control configuration, which determines the way imaging information is used in the control loop. Six categories of the visual servo control configurations and different types of controllers are briefly discussed in the following.

Visual servoing configuration

There are a few common configurations in medical robotic visual servoing applications. These are listed below, and a simple block diagram is included to describe each configuration. Note that in Figures 1 and 2, dotted lines represent non-continuous interactions, e.g. when the user specifies a target point by clicking on an image. Dashed lines and blocks represent optional functionalities that may or may not exist, e.g. pre-operative images may or may not be used in an application.

Figure 1a represents a general block diagram of image-guided medical robotic techniques which do not use visual servoing (configuration A). A common feature in these techniques is that intra-operative imaging is used to provide visual feedback to the human operator, rather than being used by the controller. The human operator may directly perform a manual procedure or may use a master–slave robot-assisted system (e.g. the da Vinci® robotic system). Examples of such configurations can be found in (6). Lee *et al.* (53) provide a systematic review of various image-guided techniques in robotic surgery.

In configuration B, depicted in Figure 1b, image-based tracking is used in order to compensate for physiological motion. In such systems, a motorized surgical tool follows the commands received from the human operator, while applying the required motion (either periodic or non-periodic) to cancel the effect of physiological motion. This feature enables the operator to work on an organ that appears to be fixed. An example of such a configuration can be found in Howe (54).

Figure 1c demonstrates another visual servoing configuration, C, in which the human operator uses a haptic device in order to control the surgical tool. At the same time, based on the obtained images and by applying visual deformation measurements and force estimation, force feedback is sent to the user through the haptic device. An example of such a configuration can be found in Kennedy *et al.* (55).

Configuration D, which covers a significant range of studies in the field of medical visual servoing, is shown in Figure 1d. In such systems, while the human is involved in the control of the robot through visualization of the surgical field and a master–slave control, intraoperative visual feedback is used in control of the operation as well. Examples of this configuration can be found in Azizian *et al.* (56) and Jayender *et al.* (57).

In the last configuration, with two variants depicted in Figure 1e, f, the clinician only has a supervisory role, e.g. in choosing the target point for the robot. The visually servoed robotic system automatically performs the procedure. The visual feedback is used to close the control loop in this configuration. An example of configuration E (Figure 1e) can be found in Jayender *et al.* (57), where the tool (a catheter) is visually servoed. An example of configuration F (Figure 1f) can be found in Belaroussi and Morel (50), where the patient position is controlled by visual servoing.

The different configurations addressed in this section are sketched in Figure 2 on a spectrum. The spectrum covers a range of configurations from manual image-guided surgery (which is the conventional method) to fully automated surgery based on visual servoing (which is the ultimate state, while still being far from reality at present). However, as seen in Table 3, the most popular configuration among the reported systems is configuration D, which is master–slave control with visual servoing. This means that visual servoing is mostly intended to close an internal control loop and leave the overall control loop to be closed through a human operator.

Visual servoing controllers

Different types of controllers have been used for visual servoing in medical applications. In motion compensation applications, predictive controllers are widely used in order to compensate for the time delays in the control loop. Predictive controllers also provide a tool for controlling systems where tracking and actuation create latency in the control loop (58). In such cases, a combination of model-based predictive estimation can also be combined with a simple controller, such as a proportional-integral-derivative (PID) controller. Robust controllers can be implemented for disturbance rejection. GPC controllers and their variations are widely used, especially when dealing with periodic physiological motions, such as heart-beat. PRC controllers have been employed in visual servoing applications by Ott *et al.* (24). Cuvillon *et al.* (59) compared GPC controllers with H_∞ controllers for visual servoing applications in minimally invasive surgery.

Real-time image processing

In order to perform visual servoing, it is required to process the acquired images in a limited time in order to detect and/or track an object in the images. A short review of the common software packages, and processing hardware used for this purpose, is included here. For an

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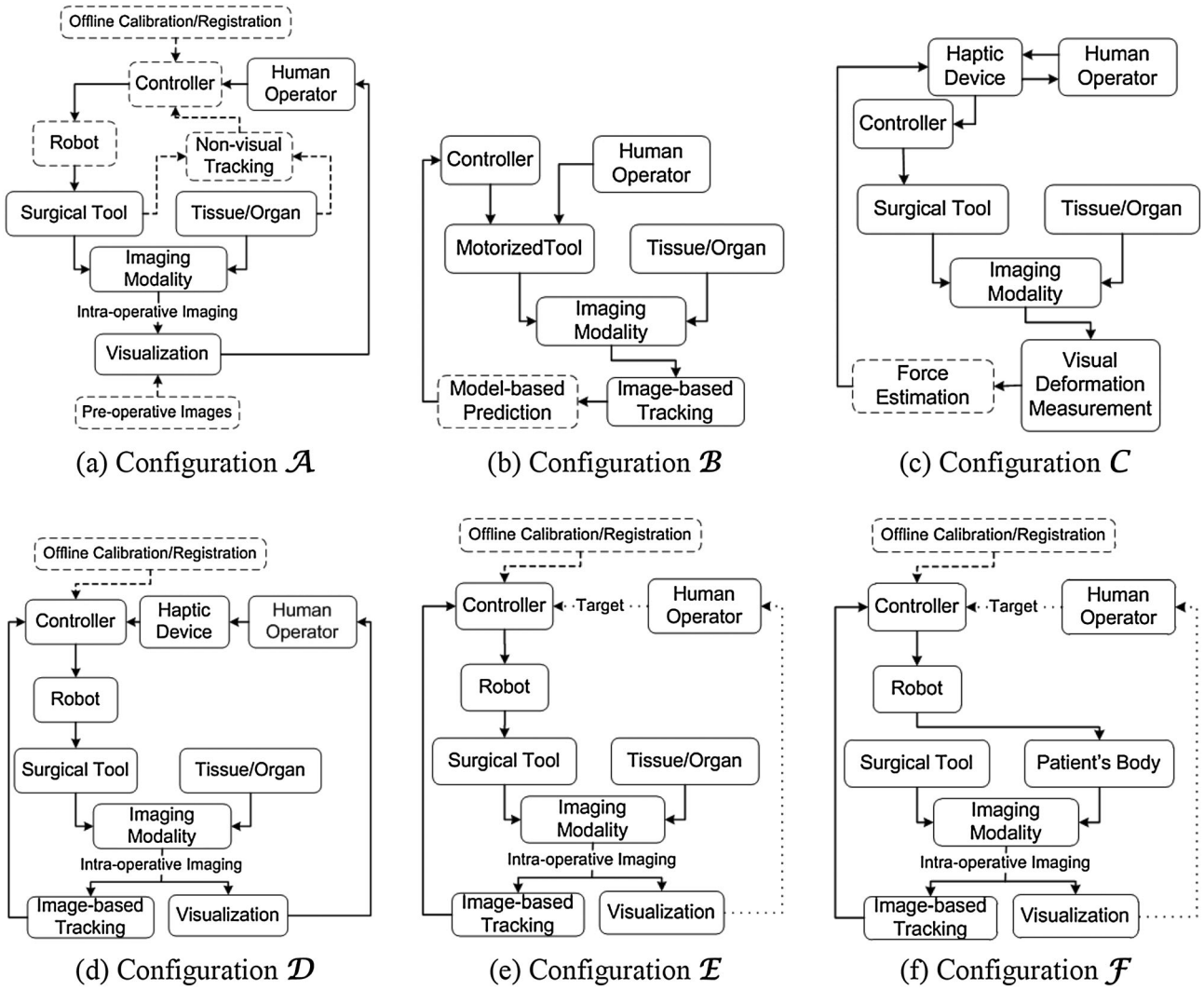


Figure 1. Configurations. (A) Image-guided surgery without visual servoing. The controller block could represent a master–slave control system. In a simplified scenario without any of the optional blocks, the surgeon controls the tool manually while looking at intra-operative images. (B) Physiological motion compensation using hand-held devices, with or without model-based prediction. (C) Master–slave control with visual force feedback. (D) Master–slave control with visual servoing. (E) Automated control with visual servoing of the surgical tool or (F) the patient's body

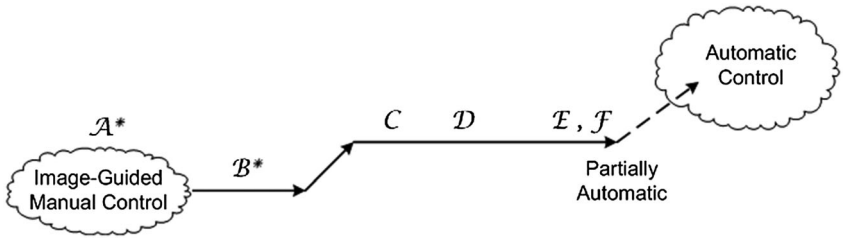


Figure 2. The spectrum of different configurations from a conventional image-guided procedure to an automated procedure. *Corresponding configuration without any of the optional functionalities (dashed blocks in Figure 1).

overview of the common detection/tracking algorithms please see Part II of the paper.

Software packages

There are several open source and commercial image-processing packages that have been broadly used in medical imaging and image-guided therapy and diagnostics. However, a main requirement for a software package to be used in visual servoing applications is the computational efficiency of the implemented algorithms. In other words, the implemented image processing algorithms should be fast enough to be used in a control loop.

Open Source Computer Vision (OpenCV)[5] is a cross-platform open-source library for real-time computer

[5]http://opencv.willowgarage.com

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vision which provides simple and efficient implementation of basic image-processing methods. Azizian *et al.* (56) used OpenCV for tracking the distal tip of a catheter in real time, which was then used in visual servoing. Reilink *et al.* (22) also used OpenCV for part of the processing required to detect the centre of the gastrointestinal tract in an endoscopic image. Bachta *et al.* (34) implemented ESM tracking for compensating for beating-heart motion using OpenCV.

Insight Segmentation and Registration Toolkit (ITK)[6] is another cross-platform open source software package developed for medical image analysis. Cleary *et al.* (60) used ITK for segmentation and tracking of surgical needles for visual servoing, using X-ray fluoroscopy images. Kahrs *et al.* (46) employed ITK for boundary detection and differentiation of cochleostomy sites.

Visualization ToolKit (VTK)[7] is an open source toolkit developed for image processing and visualization. It has been widely used by the medical imaging community and played a major role in visual servoing applications (60–64), mainly for visualization purposes and also for its capabilities in manipulating 3D image data. The Medical Imaging Interaction Toolkit (MITK)[8] is another open-source software package which combines VTK and ITK for the development of interactive medical image processing software that can potentially be used for visual servoing applications. 3D Slicer is an open-source software package that provides powerful tools for image analysis and visualization.[9]

Visual Servoing Platform (ViSP)[10] is an open source, cross-platform and modular software package that allows fast development and simulation of visual servoing applications (65). ViSP has been implemented in C++, originally under Linux (also supported in Windows and OSX). ViSP has been used by a number of researchers to develop and evaluate visual servoing algorithms with medical applications. Krupa *et al.* (63,64) and Mebarki *et al.* (61) have used ViSP for both implementation and simulation of US visual servoing, discussed in Part II of the paper.

Processing hardware

Intel Integrated Performance Primitives (IPP) is a library of multicore, highly optimized software functions for digital media/data processing. OpenCV can use Intel IPP to speed up the processing on Intel processor-based workstations. Bachta *et al.* (34) have employed Intel IPP for real-time image processing on a multi-core processing computer for visual servoing applications. Although the multi-core CPUs are becoming more and more popular for parallel processing in real-time applications, graphics-processing units (GPUs) can provide up to hundreds of parallel processing units that can be used for parallel processing. Novotny *et al.* (66) used a GPU for processing of 3D ultrasound images in visual servoing for mitral valve repair. However, there is a trade-off between the time required for data transfer and that for data processing, as GPUs provide fast parallel processing but it takes a certain amount of time to transfer the raw data to the GPU before the processing can take place. Therefore, the overall processing time might be faster on a CPU or a GPU, depending on the application. Although a GPU may have a faster processing speed, a CPU may be more efficient overall because: (a) data transfer from CPU to GPU may become a bottleneck; and (b) some algorithms may not be suitable for parallel implementation on a GPU (67). Daga *et al.* (67) provide a detailed analysis of CPU, GPU and combined CPU/GPU implementations.

[6]http://www.itk.org
[7]http://www.vtk.org
[8]http://www.mitk.org
[9]http://www.slicer.org
[10]http://www.irisa.fr/lagadic/visp/visp.html

Future directions

Rapid increase of processing power on CPUs and GPUs and the increase of data transfer rates makes it possible to transfer more data and perform complicated processing at higher rates and with lower latency. This makes it possible to run complicated image-processing and optimization algorithms in real time, and therefore to use these algorithms to develop visual servo controllers. The following three scenarios can provide better intuition. (a) A deformable registration algorithm, e.g. mutual information-based registration, is used to register an image of a deformed organ to a preoperative image or to register two consecutive images of an organ during the operation in real time. This potentially converts the registration algorithm to a real-time tracking algorithm. (b) A sophisticated segmentation algorithm, e.g. graph cuts, is implemented in real time, which makes it possible to track a tumour in US images in the presence of respiratory motion. (c) A hierarchical feature-based tracking algorithm is used to track surgical tools held by a robot in real time, and the tracking information is used for visual servoing. The possibility of using more accurate and robust image-processing algorithms implemented in real time can potentially increase the reliability of visual servoing controllers.

Towards supervised automation of medical robotics

Although a majority of the applications of visual servoing have been aimed at enabling the clinician to perform robot-assisted therapy or diagnosis, there is a trend towards making certain procedures autonomous by using visual servoing under the supervision of a human operator. Development of visual servo controllers depends greatly on the robustness of image-based detection and tracking algorithms. While there have been many developments in image processing and computer vision techniques, there are still several drawbacks in terms of

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using imaging to automate certain surgical procedures. Nageotte *et al*. (68) studied the kinematic analysis of an optimal stitching task. This can be used to automate suturing and stitching tasks through visual servoing. Non-critical repetitive surgical tasks, such as wound closure and anastomosis (connection or reconnection of tubular structures such as blood vessels, bile ducts and intestine), are good candidates for automated procedures using visual servoing techniques.

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Conflict of Interest

The authors have stated explicitly that there are no conflicts of interest in connection with this article.

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